SPE-229200-MS

AI-Driven Multi-Scale Heterogeneity Characterization of Marine Sandstone Reservoirs Based on Intelligent Thin Section Identification Technology

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This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

Abstract

This research resolves critical limitations in traditional thin section analysis of marine sandstone reservoirs through development of AI-driven intelligent identification technology based on the intelligent thin section identification system. comprehensive workflow incorporating automated multi-angle image acquisition, deep learning-based segmentation, and quantitative multi-scale characterization was established. Four representative wells exhibiting distinct sedimentary structures were analyzed: parallel bedding, cross-bedding, herringbone cross-bedding, and massive bedding. Systematic directional sampling (vertical and horizontal orientations) enabled comprehensive anisotropy characterization. Key findings include: (1) Mineral composition variations (quartz: 35-45%, feldspar: 20-30%) systematically correlate with sedimentary structures, with local enrichment reaching 60-75% in high-maturity zones; (2) Directional anisotropy analysis reveals vertical sections exhibit higher average porosity (10.5%) but reduced connectivity (0.6-0.7) compared to horizontal sections (8.5% average porosity, 0.7-0.8 connectivity); (3) AI-enhanced analysis successfully quantifies millimeter-scale grain size oscillations and compaction indices (0.1-1.4 range) previously undetectable by conventional methods; (4) Cross-scale integration establishes quantitative relationships from microscopic mineral-pore characteristics to macroscopic reservoir properties. The technology demonstrates exceptional potential for developing automated reservoir characterization systems, significantly reducing analytical subjectivity while enhancing precision and efficiency in complex marine depositional systems, providing robust foundations for future intelligent geological modeling and reservoir development optimization.

Keywords: Artificial Intelligence, Intelligent Thin Section Identification, Marine Sandstone, Reservoir Heterogeneity, Deep Learning, Multi-Scale Characterization

Introduction

Marine sandstone reservoirs serve as important hosts for oil and gas resources, representing approximately 9-17% of global oil reserves (Dou et al., 2024). The heterogeneity of these reservoirs significantly impacts hydrocarbon distribution, migration, and accumulation, directly affecting exploration and development efficiency (Bjørlykke, 2015; Liu, et al., 2019; Abitkazy, et al., 2021). Understanding and characterizing reservoir heterogeneity across multiple scales is essential for optimizing hydrocarbon recovery from marine sandstone systems.

Marine sandstone reservoirs present unique characterization challenges due to their complex depositional environments. Shoreface depositional systems exhibit intricate hydrodynamic conditions that result in widespread development of various sedimentary structures including cross-bedding, parallel bedding, and other architectural elements (Yan et al., 2010; Liu, et al., 2019). These complex internal structures create significant sedimentary heterogeneity, making fine-scale reservoir characterization particularly challenging. The rapid variation in microscopic scale characteristics within the same depositional background further complicates traditional analysis approaches, as current reservoir heterogeneity characterization techniques suffer from high subjectivity, low efficiency, and difficulties in quantification (Wang et al., 2019).

The evolution of thin section identification technology has progressed through several distinct phases (Figure 1). Traditional methods (pre-1980s) relied primarily on optical microscopy and manual interpretation. Computer-assisted analysis (1990s-2010s) introduced digital processing but maintained subjective elements. Machine learning approaches (2010s-2020) began reducing subjectivity through algorithmic methods. Recent advances in artificial intelligence, particularly visual large models such as the Segment Anything Model (SAM), have created new opportunities for geological analysis (Ren et al., 2022). These developments represent a paradigm shift toward objective, quantitative reservoir characterization methods specifically applicable to complex marine sandstone systems (Iassonov et al., 2009; Oluwadebi et al., 2019; Bjørlykke, 2015; et al., 2010; Yan et al., 2010; Liu, et al., 2019;Wang et al., 2019).

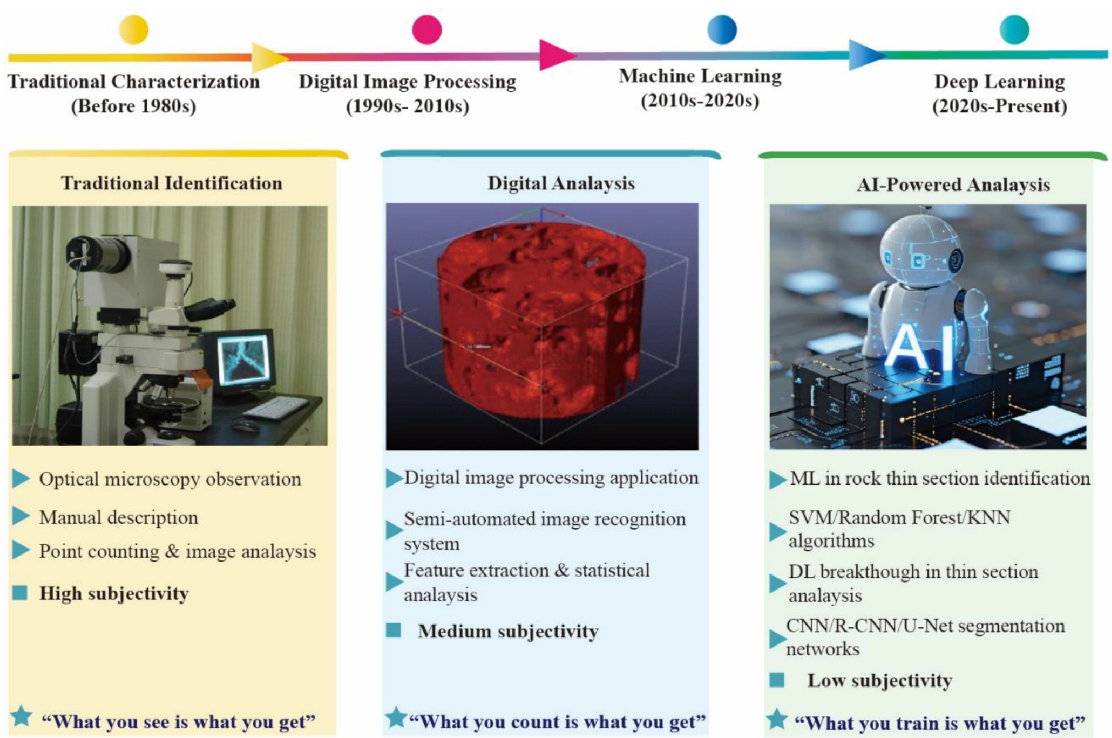

Figure 1—Evolution and development timeline of intelligent thin section identification technology in petroleum geology. The four stages are: (1) traditional optical microscopy era (pre-1980s) with manual interpretation; (2) computer-assisted digital processing period (1990s–2010s); (3) machine-learning implementation phase (2010s–2020); and (4) current AI-driven intelligent identification using vision-language foundation models such as the Segment Anything Model (SAM). The progression demonstrates increasing automation, objectivity, and quantitative capability in marine sandstone reservoir characterization.

Current reservoir heterogeneity characterization techniques can be broadly categorized into visual imaging characterization methods and indirect numerical measurement approaches (Petrophysical parameter determination; Figure 2). Visual imaging methods include cast thin sections, field emission scanning electron microscopy (FE-SEM), quantitative evaluation of minerals by scanning electron microscopy (QEMScan), environmental scanning electron microscopy (ESEM), atomic force microscopy (AFM), micro-nano X-ray computed tomography (X-CT), and focused ion beam scanning electron microscopy (FIB-SEM) (Abitkazy et al., 2024;2025). Petrophysical parameter determination methods primarily include conventional mercury intrusion capillary pressure (MICP; Su et al., 2023), constant-rate mercury intrusion, gas adsorption, nuclear magnetic resonance (NMR), small-angle scattering (SAS), and log-based calculation methods (Al-Mahrooqi et al., 2003). Each characterization technique has its advantages and limitations, with different measurement ranges and capabilities.

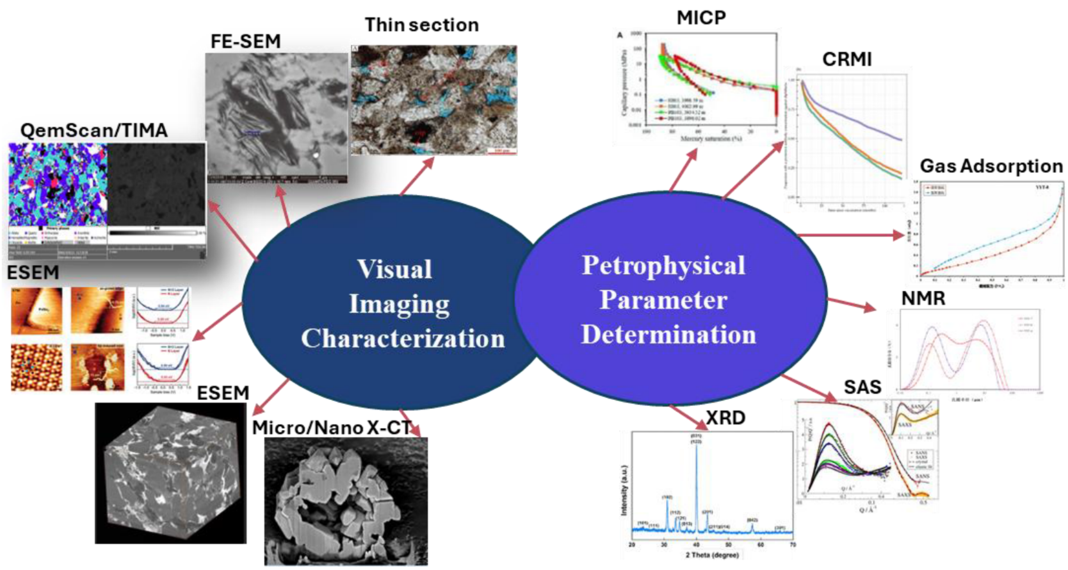

Figure 2—Comprehensive classification of reservoir heterogeneity characterization methods showing technical capabilities and measurement ranges (modified after Al-Mahrooqi et al., 2003; Ren et al., 2022;Su et al., 2023; Abitkazy, et al., 2024;2025). Visual imaging methods including cast thin sections, FE-SEM, QEMScan, ESEM, AFM, X-CT, and FIB-SEM with nanometer to millimeter scale resolution; Indirect petrophysical measurement approaches including MICP, gas adsorption, NMR, and log-based calculations spanning pore to reservoir scale. Each method's advantages, limitations, and optimal application scenarios are indicated for marine sandstone reservoir evaluation.

This research aims to develop and implement an AI-driven intelligent thin section identification workflow specifically designed for marine sandstone reservoir heterogeneity characterization, establishing automated quantitative analysis capabilities for complex marine depositional structures and achieving multi-scale characterization from microscopic minerals to macroscopic reservoir properties.

Materials and Methods

Sample Preparation and Data Acquisition

Marine Sandstone Sample Selection and Preparation. This study focuses on lower shoreface depositional environments within marine sandstone reservoirs, which are characterized by complex hydrodynamic conditions and diverse sedimentary structures. The research area represents a typical marine shelf setting where fluctuating wave energy and tidal currents create heterogeneous depositional architectures that significantly impact reservoir properties.

Four representative wells were systematically selected based on their comprehensive representation of different sedimentary sequences within the same lower shoreface depositional background (Figure 3a). Selection criteria prioritized wells exhibiting spatial distribution of multiple bedding types, heterogeneity feature representation, and depositional continuity from the same stratigraphic horizon to ensure comparable depositional conditions while capturing structural variations.

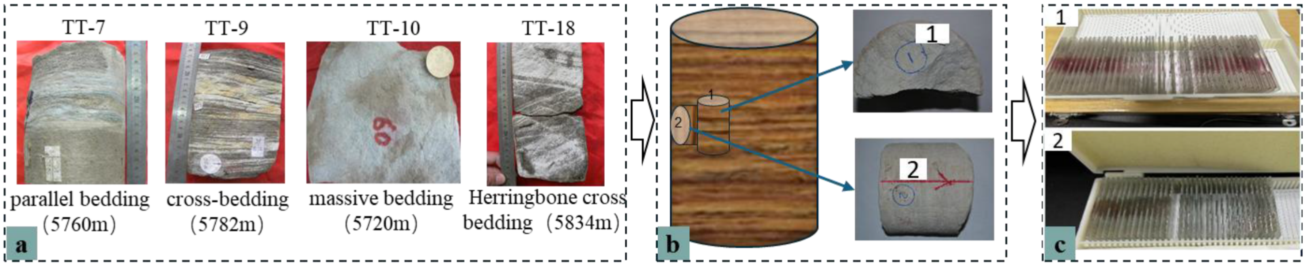

Figure 3—Sample processing workflow. (a) Core-sample selection from four representative wells illustrating distinct sedimentary sequences; (b) Systematic preparation of cast thin sections in vertical and horizontal orientations; (c) Standardized thin-section dimensions and quality control.

The selected core samples represent distinctive sedimentary sequences:

1. Well TT-7 (5760m depth) - Parallel bedding with consistent lamination patterns.
2. Well TT-9 (5782m depth) - Cross-bedding with well-developed trough structures.
3. Well TT-18 (5834m depth) - Herringbone cross bedding with characteristic curved lamination patterns
4. Well TT-10 (5720m depth) – Massive bedding (structureless) with minimal internal structure.

Systematic Sample Preparation Protocol. To achieve precise characterization of reservoir heterogeneity features, a comprehensive sample preparation protocol was implemented following strict technical specifications for marine sandstone analysis.

For fine-scale delineation of reservoir heterogeneity characteristics, cast thin sections were systematically prepared from identical core samples in both vertical and horizontal orientations (Figure 3b). This multi-directional sampling approach enables multi-dimensional analysis of pore structure and mineral composition spatial heterogeneity, providing microscopic-scale evidence chains for quantitative reservoir heterogeneity evaluation. Cast thin sections were systematically prepared from identical core samples at the same depth (same point) in both vertical and horizontal orientations to better characterize heterogeneity (Figure 3b). This multi-directional sampling approach from the same location enables comprehensive analysis of anisotropic characteristics and provides microscopic-scale evidence for quantitative reservoir heterogeneity evaluation.

All thin sections were prepared in both vertical and horizontal orientations according to standardized dimensions optimized for integrated intelligent characterization as 23-26mm width, ≤76mm length standard size systems.

Intelligent Thin Section Identification Workflow

This research implements an intelligent thin section identification workflow based on the CoreSAM (Core Segment Anything Model) system, following the comprehensive framework described by Ren et al. (2025) for intelligent sandstone rock structure evaluation using visual large models (Figure 4). The workflow establishes a comprehensive "sample submission-intelligent identification-report generation" processing pipeline specifically adapted for marine sandstone reservoir characterization.

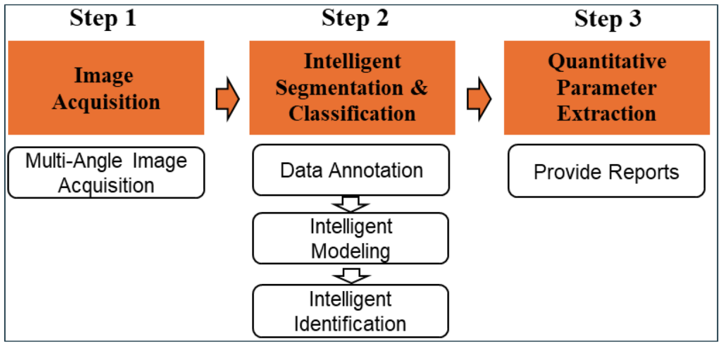

Figure 4—Intelligent thin section identification workflow (modified after Ren et al., 2025).

Step 1: Multi-Angle Image Acquisition. Using automated thin section scanners, full-field images are obtained revealing significant heterogeneity in sedimentary characteristics (Figure 5a). The imaging protocol captures one plane-polarized light image and six cross-polarized light images at different angles (0°, 15°, 30°, 45°, 60°, 75°), ensuring comprehensive coverage of optical properties for marine sandstone samples (Figure 5b).

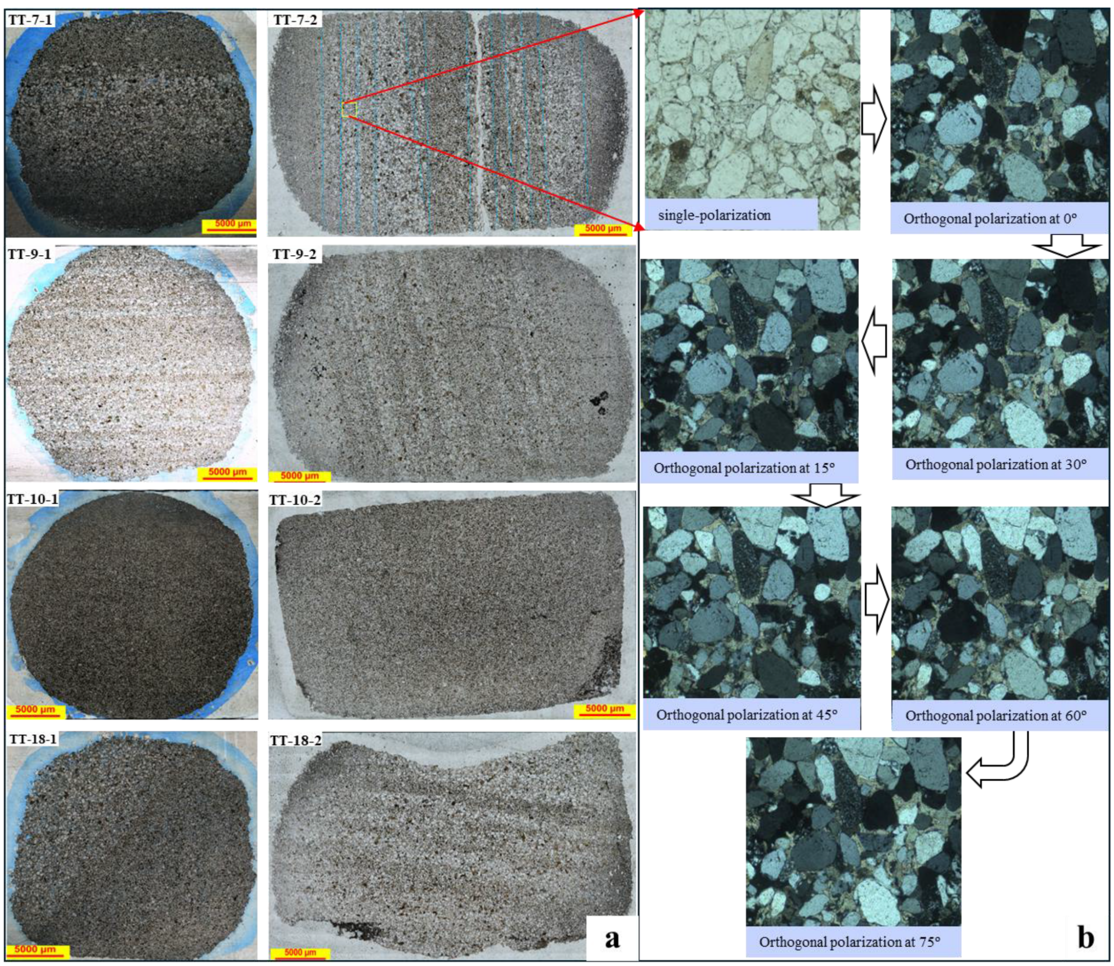

Figure 5—Standardized multi-angle imaging protocol for marine sandstone thin section analysis: (a) Full-field plane-polarized light images (Scale bar = 5000 μm) revealing macroscopic sedimentary heterogeneity and structural patterns in samples from different depths; (b) Cross-polarized light image series at systematic angles (0°, 15°, 30°, 45°, 60°, 75°) enabling comprehensive mineral optical property characterization and automated identification of quartz (colorless to gray), feldspar (twinned crystals), and accessory minerals. This imaging protocol provides standardized input for CoreSAM deep learning analysis.

Step 2: Intelligent Segmentation and Classification. The CoreSAM system processes acquired images through systematic data annotation followed by deep learning modeling. The process integrates multiple analytical components including pore structure analysis, mineral composition identification, grain segmentation, and grain contact relationship characterization specific to marine sandstone reservoir characteristics.

Step 3: Quantitative Parameter Extraction. Automated extraction and generation of comprehensive analytical reports containing quantitative parameters including:

- Grain size distribution analysis with statistical characterization
- Pore structure analysis including porosity, pore connectivity, and pore size distribution
- Mineral composition identification and quantification of major and minor components
- Sorting coefficient and roundness quantification
- Contact relationship and cementation type classification

All quantitative parameters reported in this study (including mineral composition, pore structure, compaction indices, and areal porosity distributions) were extracted by CoreSAM imaging of the resin-impregnated thin sections prepared in both vertical and horizontal orientations from four wells (n = 8).

Results and Discussion

Marine Sandstone Microscopic Heterogeneity Characterization

The intelligent thin section identification system successfully characterized the complex heterogeneity features of marine sandstone reservoirs across different sedimentary sequences Petrographic analysis indicates that marine sandstone samples are predominantly composed of medium-fine to very fine-grained feldspathic lithic sandstones with moderate weathering degrees, typical of marine shelf depositional environments.

Marine Sandstone Grain Size Distribution Analysis. Grain size analysis of marine sandstone samples reveals complex distribution patterns characteristic of marine depositional systems. Grain size distributions reveal four distinct hydraulic microenvironments characteristic of nearshore depositional facies:

Parallel laminations (TT-7-1/TT-7-2) exhibit 60-120 μm fine-grained sand with sorting coefficients less than 0.7, indicating low-energy swash zone conditions. Cross-stratified units (TT-9-1/TT-9-2) show 0.25-0.50 mm coarse sand peaks with moderate sorting (0.5-0.9 Φ), reflecting surf-backwash alternation. Massive intervals (TT-10-1/TT-10-2) display broad distributions (0.18-0.50 mm, σ 0.7-1.2 Φ) from bioturbation mixing. Herringbone cross-bedding laminations (TT-18-1/TT-18-2) concentrate in 0.12-0.25 mm fine sand with bidirectional flow signatures. Matrix content remains below 3%, with maximum grain size reaching 1.6 mm.

The AI-assisted analysis reveals microscale heterogeneities within apparently uniform parallel laminations. This approach achieves unprecedented precision in reservoir characterization. The intelligent system successfully captures previously undetected multi-scale grain size oscillations within apparently uniform parallel laminations, demonstrating the superior analytical capabilities of the CoreSAM technology over traditional methods.

Marine Sandstone Mineral Composition Analysis. Automated mineral identification reveals systematic spatial variations characteristic of marine sandstone systems:

Framework composition ranges: Quartz 35-45%, feldspar 20-30%, with localized enrichment zones reaching up to 60-75% quartz in high-maturity intervals

Cement distribution: Dolomite cement (5-10%) in parallel-laminated samples, chlorite coatings (8-15%) in cross-stratified intervals

Diagenetic evolution: Progressive maturation from dolomite-dominated → intermediate tuffaceous → chlorite-siderite assemblages

The AI system successfully quantifies these mineralogical heterogeneities with enhanced precision and reduced analytical bias. Automated mineral identification achieves an average accuracy of 95%, as validated against QEMScan reference measurements.

Marine Sandstone Pore Structure Characteristics. Pore structure analysis reveals the distinctive pore system characteristics of marine sandstone reservoirs (Figure 6). Sedimentary structures control reservoir space distribution in systematic patterns:

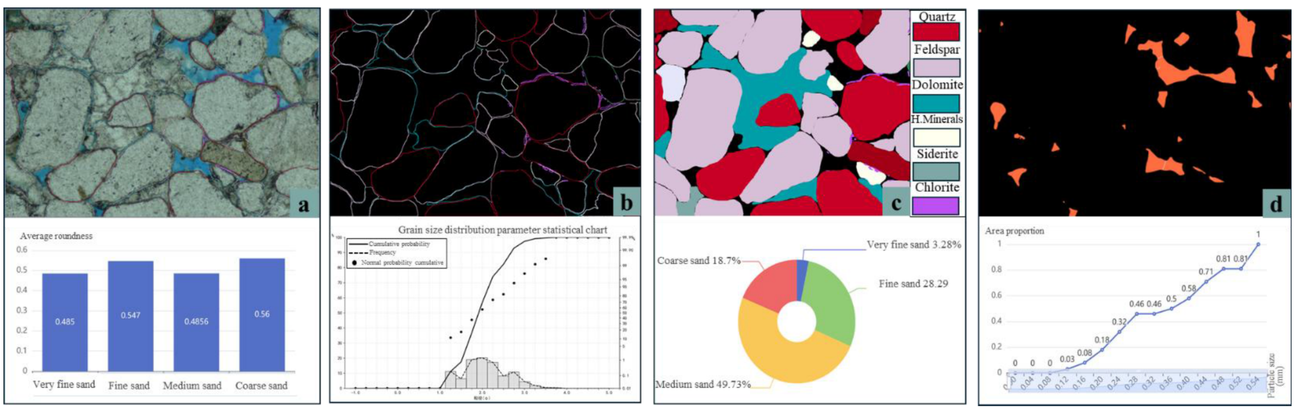

Figure 6—Comprehensive AI-driven thin section analysis results demonstrating multi-parameter characterization capabilities (a) Automated rock fabric analysis revealing grain boundary relationships and packing arrangements; (b) High-resolution grain size distribution analysis with statistical parameters (n=500+ measurements per sample); (c) Quantitative mineral phase identification with >95% accuracy; (d) Comprehensive pore structure characterization including connectivity analysis and pore-throat size distributions.

Parallel laminations (TT-7-1/TT-7-2) show 50-90 μm pores, 8-12 μm throats, with porosity <3% (low-porosity fine-throat type). Cross-stratified samples (TT-9-1/TT-9-2) exhibit 70-160 μm pores, 10-25 μm throats, with 3-6% porosity. Massive intervals TT-10-1/TT-10-2) display >180 μm pores but <5 μm throats with 0-2% porosity from compaction effects. Herringbone cross-bedding laminations (TT-18-1/TT-18-2) achieve optimal 120-200 μm pores, 15-35 μm throats, 6-10% porosity with high connectivity networks.

The digital analysis enables precise multi-scale pore architecture characterization, providing quantitative frameworks for reservoir quality assessment in marine sandstone systems.

Directional Anisotropy in Marine Sandstone Reservoirs

Compositional Complexity Comparison. Comparative analysis between vertical and horizontal samples from identical depths reveals significant anisotropic contrasts in compositional characteristics within the same marine depositional background. All quantitative parameters reported in this analysis were extracted through CoreSAM imaging of resin-impregnated thin sections prepared in both orientations.

Mineral Distribution Patterns: Vertical sections exhibit pronounced compositional variations with quartz content coefficient of variation (0.15-0.25) significantly exceeding horizontal sections (0.08-0.15), indicating stronger vertical compositional gradients in marine shoreface environments.

Vertical mineral heterogeneity is primarily controlled by bedding types (Figure 7a, 7c, 7e): Parallel bedding (TT-7-1) displays feldspar enrichment-depletion layers precisely superimposed with millimeter-scale heavy mineral bands, reflecting periodic protection of labile components during traction flow aggradation. Cross-bedding (TT-9-1) shows selective feldspar dissolution in upper foresets while preservation at bases, with heavy minerals concentrated at bottom interfaces, revealing dissolution-precipitation coupling driven by high-energy swash events through interfacial fluid activity. Massive bedding (TT-10-1) exhibits vertically uniform minerals with only localized minor feldspar dissolution patches, indicating limited dissolution under restricted interfacial fluid disturbance. Herringbone cross-bedding (TT-18-1) demonstrates mirror-image feldspar variations between couplet layers synchronized with fine-grained heavy mineral bundles, where bidirectional flows periodically alter oxidation-reduction interfaces controlling dissolution-preservation equilibrium.

Horizontally, all bedding types display mineral continuity (Figure 7b, 7d, 7f): TT-7-2, TT-9-2, TT-10-2, and TT-18-2 show feldspar and quartz extending continuously along lamination strikes at centimeter scales, heavy mineral bands maintain lateral traceability, and dissolution differences are laterally homogenized. Overall heterogeneity decreases by approximately one order of magnitude compared to vertical profiles.

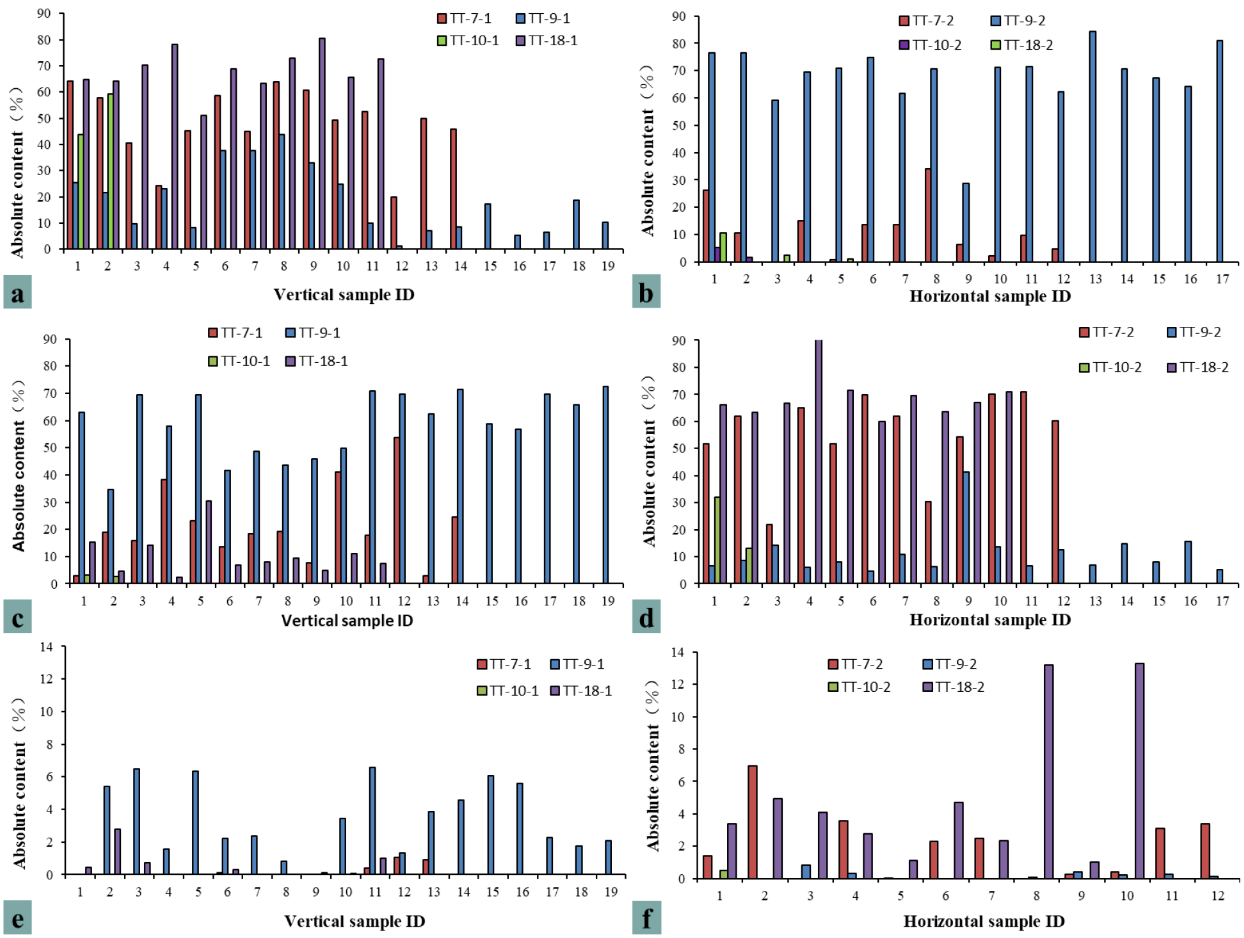

Figure 7—Intra-sample directional anisotropy of framework minerals (quartz, feldspar, heavy minerals) in marine sandstones: paired vertical (a, c, e)and horizontal (b, d, f) thin-section analyses for each of eight samples from four sedimentary-architecture-controlled facies (parallel-, cross-, herringbone-cross-, and massive-bedding, wells TT-7, TT-9, TT-10,TT-18), quantified by CoreSAM imaging of resin-impregnated sections.

This directional contrast demonstrates that sedimentary structures control mineral heterogeneity through fluid action orientation—vertical profiles dominated by periodic diagenetic modification, horizontal profiles homogenized through mineral lateral continuity. Quartz serves as an 'inert ruler' for bedding patterns and fluid activity intensity due to its high chemical stability, further confirming that bedding types control marine sandstone reservoir mineral heterogeneity evolution through interfacial fluid chemical gradients.

Clay Matrix Distribution: Vertical clay content heterogeneity (3-8%) couples strictly with bedding types (Figure 8a, 8c, 8e): Parallel laminations (TT-7-1) show dolomite centimeter-scale bands synchronized with grain size rhythms, tuffaceous materials enriched in fine-grained segments, chlorite films following lamination undulations. Cross-bedding (TT-9-1) displays dolomite patches in bottomsets indicating rapid post-high-energy precipitation, with tuffaceous materials replaced by chlorite in foreset tops. Massive intervals (TT-10-1) exhibit sparse dolomite, dispersed tuffaceous materials, and uniform chlorite films reflecting weak cementation environments. Herringbone cross-bedding (TT-18-1) shows bidirectional "mirror-image" dolomite variations with bundled tuffaceous enrichment and chlorite thickening at couplet interfaces.

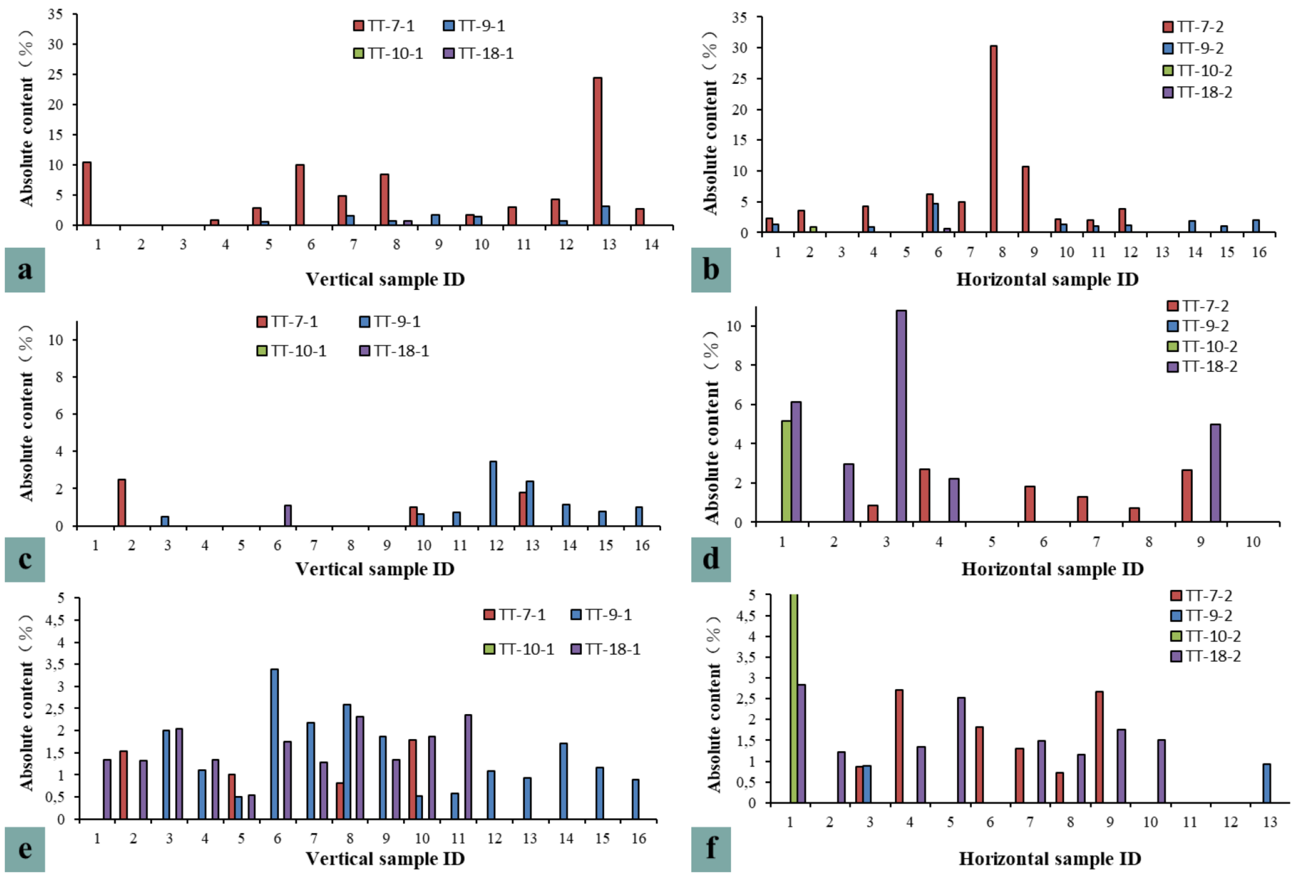

Figure 8—Intra-sample directional anisotropy of diagenetic matrix minerals (dolomite, tuffaceous material, chlorite) in marine sandstone reservoirs: paired vertical (a, c, e) and horizontal (b, d, f) thin-section analyses for each of eight samples from four sedimentary-architecture-controlled facies (parallel-, cross-, herringbone-cross-, and massive-bedding, four wells), quantified by CoreSAM imaging of resin-impregnated sections.

Horizontally, clay matrices demonstrate high continuity (4-6% variation) (Figure 8b, 8d, 8f): TT-7-2, TT-9-2, TT-10-2, and TT-18-2 show dolomite forming laterally continuous thin layers, tuffaceous materials and chlorite constituting parallel fine laminae with significantly reduced heterogeneity. Volcanogenic tuffaceous materials show lateral homogenization without vertical rhythmic characteristics. This distribution pattern confirms directional control of sedimentary structures on diagenetic heterogeneity—vertical coupling with bedding and diagenetic fluid differentiation, horizontal continuity with heterogeneity smoothing, reflecting periodic diagenetic processes.

Structural Diversity Comparison. **Grain Contact Relationships:** Vertical sections display variable contact types (point contacts: 15-30%) reflecting internal bedding architecture, while horizontal sections show consistent patterns (18-25%) indicating uniform lateral stress distribution (Figure 9a).

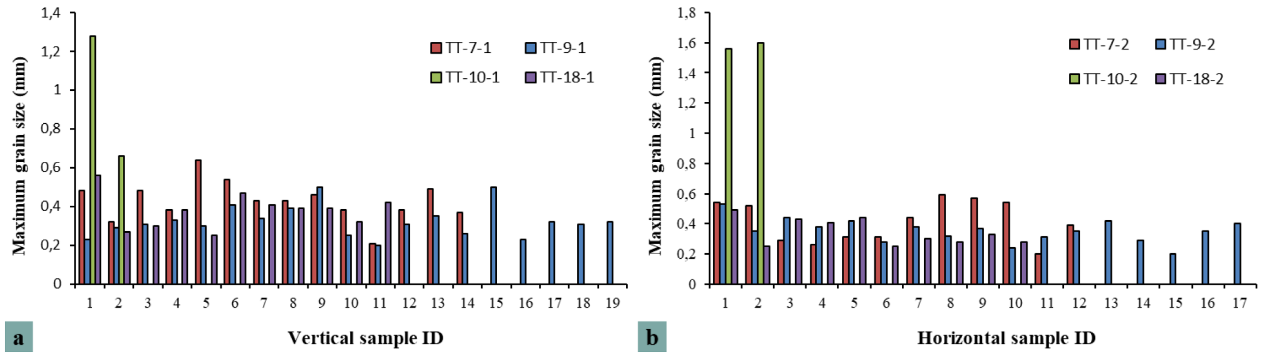

Figure 9—Multi-scale hydrodynamic signatures in lower-shoreface deposits revealed by intra-sample grain-size heterogeneity: paired vertical (a) and horizontal (b) thin-section measurements of maximum grain size for each of eight samples from four sedimentary-architecture-controlled facies (parallel-, cross-, herringbone-cross-, and massive-bedding, four wells), quantified by CoreSAM imaging of resin-impregnated sections.

Multi-scale hydrodynamic signatures revealed through High-resolution grain size analysis of vertical and horizontal thin sections reveals multi-scale hydrodynamic-diagenetic coupling systems in lower shoreface environments. Vertical samples (TT-7-1, TT-9-1, TT-10-1, TT-18-1) exhibit millimeter-scale "sawtooth-stepped" rhythms with Φ values oscillating between 2.1-3.0 in 3-5 mm cycles, sorting $\sigma<0.7$, kurtosis 4-12, reflecting tidal-wave alternation forming swash couplets. Argillaceous laminae (2-6% matrix) and pyrite microcrystals mark storm waning suspension settling, while carbonate cementation forms discontinuous millimeter-scale hardgrounds along lamination interfaces preserving high-frequency grading signals.

Horizontal samples (TT-7-2, TT-9-2, TT-10-2, TT-18-2) show coarser overall grain sizes by 0.1-0.3 Φ with σ 0.5-1.2 Φ and bimodal/broad peaks indicating scour-backflow repetition (Figure 9b). This directional contrast reflects energy-controlled depositional architectures where vertical heterogeneity preserves tidal rhythmicity while horizontal continuity records lateral transport consistency.

Grain Fabric Anisotropy: Vertical fabric discloses tidally-driven grain orientation, while lateral fabric tracks depositional continuity (Figure 9).

- Parallel bedding shows high-frequency tidal laminae; vertically, grains (max 2.1 mm) align 15–25° imbricate to flow, with diagenetic cement sharpening contacts. Horizontally, grains (max 2.0 mm) are mixed and homogenized.
- Cross-bedding exhibits rhythmic fine laminae (max 0.9 mm); foresets record 45–60° directional shift upward, tracing waning flow. Laterally, coarser lags (max 1.7 mm) with broad sorting reflect surf-zone scour and weak cement.
- Herringbone cross-bedding carries bidirectional coarse–fine alternations (max 1.3 mm vertically, 2.4 mm laterally); vertical anisotropy 0.6–0.8. Mud drapes persist laterally yet are diagenetically blurred.
- Massive structureless sandstones are texturally homogeneous, yet contain scattered outsized grains (max 1.5 mm vertical, 1.6 mm lateral) within an otherwise uniform matrix. Sparse, patchy cementation and bimodal peaks reflect local winnowing and brief high-energy pulses rather than biogenic reworking; grain orientations remain nearly isotropic (anisotropy <0.3), preserving primary depositional uniformity.

This directional fabric encodes energy flux: vertical heterogeneity archives tidal rhythm, lateral continuity captures transport uniformity. AI-driven grain-size mapping resolves millimetre-scale patch genesis, refining digital-core models.

Pore Structure Anisotropy: Cross-directional analysis demonstrates systematic connectivity differences: vertical samples show lower connectivity ratios (0.6-0.7) while horizontal samples exhibit better connectivity (0.7-0.8), reflecting depositional lamination influence on pore network development (Figure 10a, 10c, 10eef).

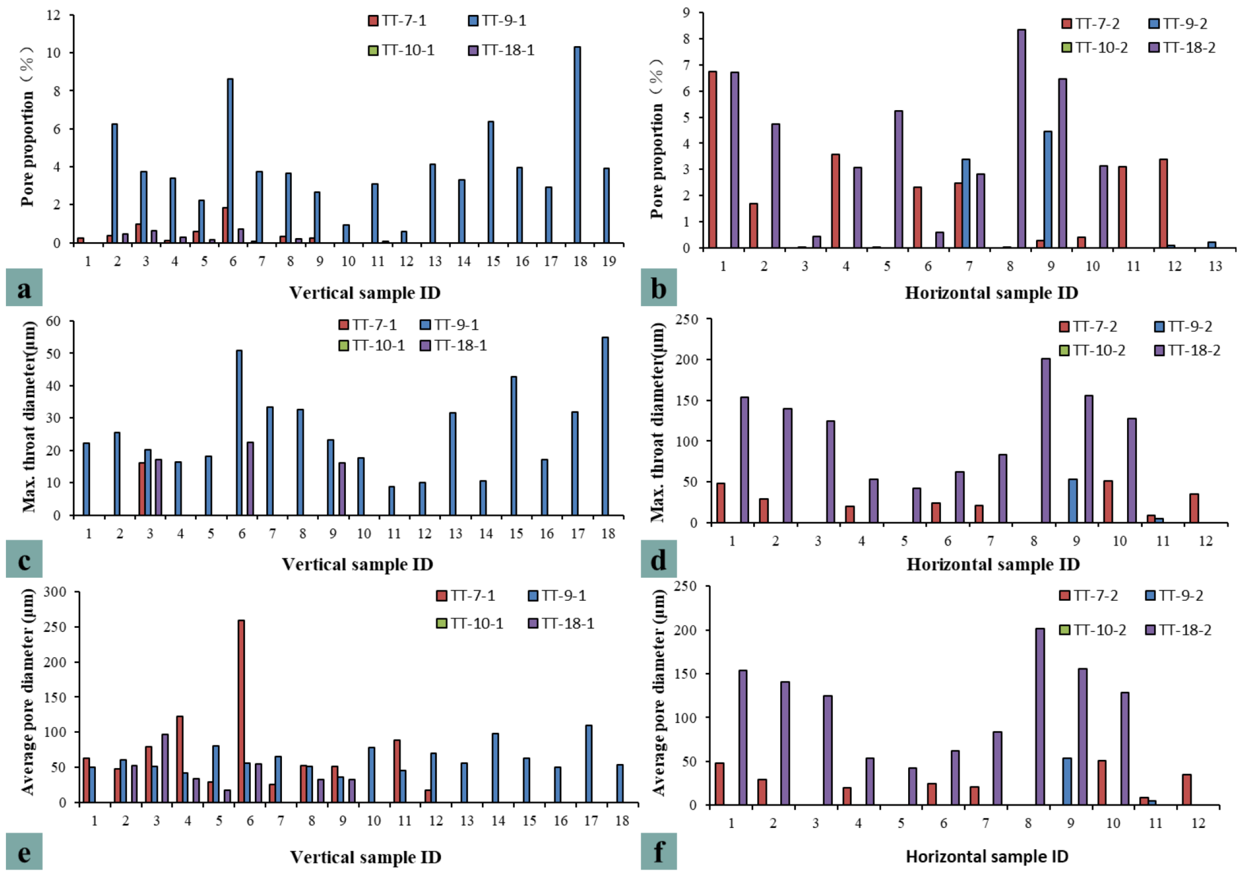

Figure 10—Comprehensive pore network architecture analysis demonstrating depositional control on reservoir quality heterogeneity: paired vertical (a, c, e) and horizontal (b, d, f) distributions of intergranular porosity, maximum throat diameter and average pore diameter in lower-shoreface deposits (parallel-, cross-, herringbone-cross-, and massive-bedding, four wells), quantified by CoreSAM imaging of resin-impregnated sections.

Intergranular pore development varies significantly vertically, indicating alternating high-permeability layers and low-permeability barriers corresponding to grain size rhythms and diagenetic differentiation. Maximum throat diameter vertical variations reflect grain sorting control and cement distribution, with concentrated large-throat intervals forming preferential flow pathways affecting waterflooding efficiency.

Horizontally, throat diameter differences reflect paleocurrent intensity variations and bioturbation effects on connectivity (Figure 10b, 10d, 10f). Mean pore diameter shows vertical reduction corresponding to burial-compaction sequences, while horizontal variations relate to depositional energy belts. These anisotropic characteristics indicate that vertical heterogeneity controls fluid vertical migration barriers/ pathways, while horizontal variations may form reservoir boundaries affecting sweep efficiency.

The systematic vertical-horizontal compaction contrast demonstrates that sedimentary structures control pore structure heterogeneity through directional fluid flow patterns—vertical profiles dominated by periodic diagenetic modification creating flow barriers, horizontal profiles showing enhanced connectivity along bedding planes facilitating lateral fluid migration.

Multi-Scale Characterization and Controlling Factors

Depositional Process Control on Multi-Scale Heterogeneity. Marine sandstone reservoir heterogeneity is fundamentally controlled by varying hydrodynamic conditions during deposition. The intelligent characterization system reveals systematic relationships between depositional energy and reservoir properties across multiple scales.

High-Energy Marine Conditions (Cross-bedding):

- Enhanced grain sorting characterized by improved sorting coefficients (0.4-0.6)
- Increased grain roundness values (0.6-0.8) indicating extended transport
- Coarser grain sizes (0.5-1.2mm) reflecting high-energy wave action
- Complex internal architecture with variable dip angles and grain fabric

Moderate-Energy Marine Conditions (Parallel Bedding,):

- Intermediate sorting characteristics with moderate coefficients (0.5-0.7)
- Consistent grain orientation patterns indicating steady flow conditions
- Uniform lamination thickness reflecting stable depositional energy
- Enhanced lateral continuity of reservoir properties

Variable-Energy Marine Conditions (Herringbone Cross-Bedding):

- Complex sorting patterns with high variability (coefficient: 0.6-0.9)
- Mixed grain populations indicating fluctuating energy conditions
- Curved lamination patterns reflecting oscillatory flow influence
- Intermediate heterogeneity characteristics

Low-Energy Marine Conditions (Massive Bedding):

- Poor sorting characteristics due to bioturbation mixing
- Random grain orientation from biological reworking
- Homogeneous appearance masking subtle depositional variations
- Reduced primary depositional heterogeneity

Diagenetic Process Control and Quantitative Characterization. Post-depositional diagenetic processes significantly modify primary depositional heterogeneity, requiring quantitative characterization approaches for accurate reservoir evaluation (Morad et al., 2010).

Compaction Effects and Quantitative Assessment: Compaction intensity quantification through grain contact relationships reveals systematic directional variations across different bedding types (Figure 11).

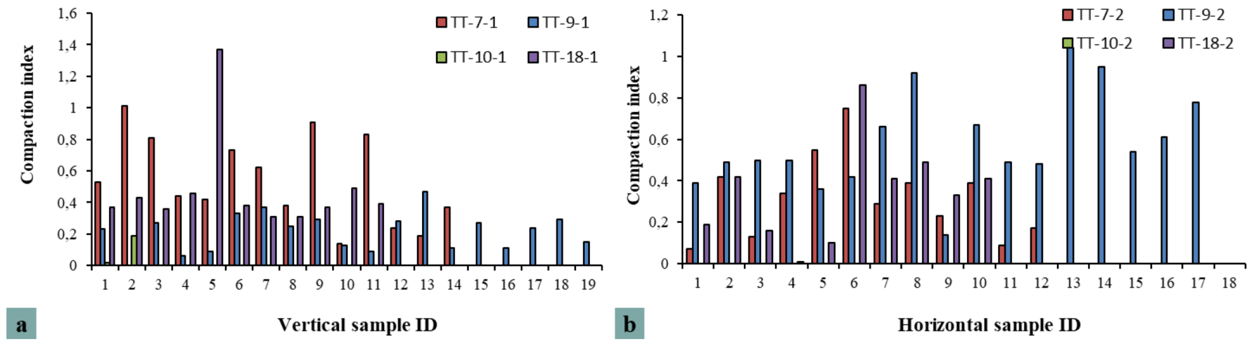

Figure 11—Quantitative compaction index variations demonstrating sedimentary control on mechanical diagenesis. (a) Vertical compaction index variations (0.1-1.4 range) controlled by energy stratification and rhythmic deposition; (b) Horizontal compaction index distributions showing lateral facies transitions and selective diagenetic modifications. Different colors represent different samples within the same orientation (vertical or horizontal direction) across four architectural facies (parallel-, cross-, herringbone-cross-, and massive-bedding), quantified by CoreSAM imaging of resin-impregnated sections (n = 8 samples, 4 wells, two orientations per well).

Vertical analysis demonstrates significant heterogeneity patterns (Figure 9a): Parallel laminated samples (TT-7-1) exhibit highly fluctuating compaction indices (0.2-1.4) reflecting periodic tidal-wave energy variations and grain size rhythms. Cross-bedded samples (TT-9-1) display moderate and relatively stable indices (0.3-0.8) indicating consistent high-energy depositional conditions with uniform compaction. Massive bioturbated samples (TT-10-1) maintain consistently low compaction levels (0.1-0.5) due to early carbonate cementation protection against mechanical compaction. Herringbone cross-bedded samples (TT-18-1) show distinct peaks (0.4-1.2) corresponding to bidirectional flow energy fluctuations and clay drape compaction enhancement.

Horizontal sections demonstrate contrasting patterns (Figure 10b): Parallel laminated samples (TT-7-2) show minimal index variations (0.3-0.6) reflecting uniform lateral stress distribution along lamination strike. Cross-bedded samples (TT-9-2) maintain consistently moderate indices (0.4-0.7) with lateral stability. Massive samples (TT-10-2) display localized high-value zones (0.2-0.9) related to differential bioturbation intensity and heterogeneous cementation. Herringbone cross-bedded samples (TT-18-2) exhibit relatively smooth variations (0.3-0.8) along strike continuity.

The AI quantification demonstrates the potential to directly correlate microscopic pore heterogeneity with sedimentary structural patterns, providing high-resolution criteria for hierarchical marine sandstone reservoir evaluation. The systematic vertical-horizontal compaction contrast demonstrates that burial-controlled vertical compaction primarily reflects energy stratification and rhythmic deposition, while horizontal variations indicate lateral facies transitions and selective diagenetic modification patterns.

Cementation Effects and Quantitative Parameters: Areal porosity distribution analysis provides quantitative characterization of cementation effects across directional and bedding-type variations.

Vertical porosity patterns (Figure 12a) reveal distinct cementation-controlled variations: Parallel laminated samples (TT-7V) demonstrate rhythmic porosity oscillations (2-12%) synchronized with grain size variations and carbonate cement precipitation cycles. Cross-bedded samples (TT-9V) show systematic porosity reduction from bottomset to topset (8-15% to 3-8%) reflecting progressive burial and cementation intensity. Massive bioturbated samples (TT-10V) exhibit patchy porosity distribution (1-10%) corresponding to bioturbation-controlled cementation heterogeneity. Herringbone cross-bedded samples (TT-18V) display bidirectional porosity couplets (4-14%) reflecting alternating oxidation-reduction diagenetic environments.

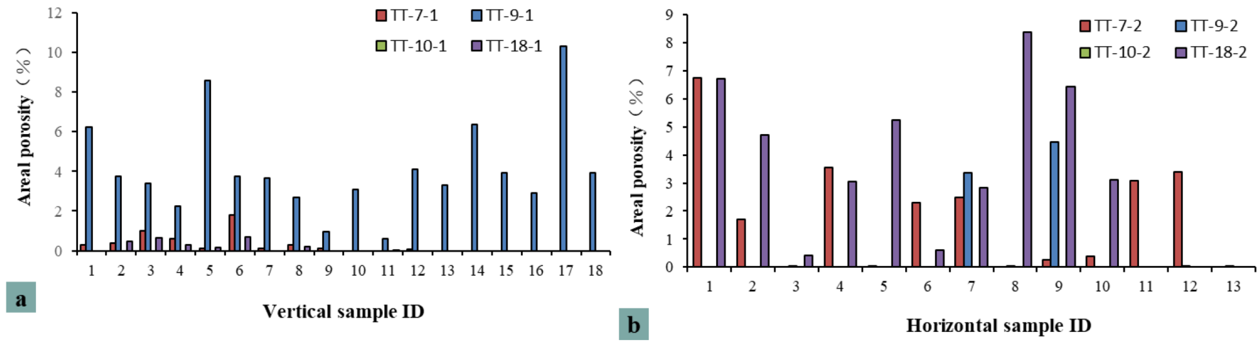

Figure 12—Areal porosity distribution analysis revealing cementation-controlled heterogeneity: (a) Vertical areal porosity oscillations (2-12% range) synchronized with grain size variations and carbonate cement precipitation cycles, demonstrating rhythmic diagenetic modification in tidal-influenced marine environments; (b) Horizontal areal porosity continuity (6-10% range) showing minimal lateral variations and enhanced connectivity along bedding planes. Cementation intensity classification: intensive (<5% porosity), moderate (5-12%), and weak (>12%) based on porosity-structure relationships. Results quantify reservoir-quality trends across four architectural facies (parallel-, cross-, herringbone-cross-, and massive-bedding), quantified by CoreSAM imaging of resin-impregnated sections (n = 8 samples, 4 wells, two orientations each).

Horizontal porosity characteristics (Figure 12b) demonstrate lateral continuity patterns: Parallel laminated samples (TT-7H) maintain relatively uniform porosity (6-10%) along strike with minimal lateral variations. Cross-bedded samples (TT-9H) show gradual lateral porosity changes (5-12%) related to paleocurrent intensity variations. Massive samples (TT-10H) exhibit localized porosity clusters (2-15%) corresponding to bioturbation intensity zones. Herringbone cross-bedded samples (TT-18H) display laterally continuous porosity bands (7-13%) following bedding geometry.

Cementation intensity classification based on porosity-structure relationships:

- Intensive cementation zones: Areal porosity <5%, dominantly sutured grain contacts, extensive carbonate overgrowth
- Moderate cementation zones: Areal porosity 5-12%, mixed contact types, localized cement precipitation
- Weak cementation zones: Areal porosity >12%, predominantly point contacts, minimal cement development

The integrated porosity-compaction analysis reveals that sedimentary structures control cementation spatial distribution patterns through diagenetic fluid pathway control—vertical profiles reflecting periodic diagenetic differentiation, horizontal profiles showing continuous cementation environments, providing quantitative frameworks for reservoir quality prediction and development optimization in marine sandstone systems.

Multi-Scale Integration: "Microscopic Minerals-Pores → Macroscopic Reservoir Properties". **Cross-Scale Lithofacies Classification Methodology:** The innovative integration of AI-enhanced microscopic characterization with macroscopic reservoir attributes provides a framework for establishing quantitative relationships across multiple scales. The cross-scale lithofacies classification methodology from "microscopic mineral-pore" to "macroscopic reservoir properties" provides multi-dimensional theoretical support for geological modeling, enabling efficient reservoir development(Figure 13).

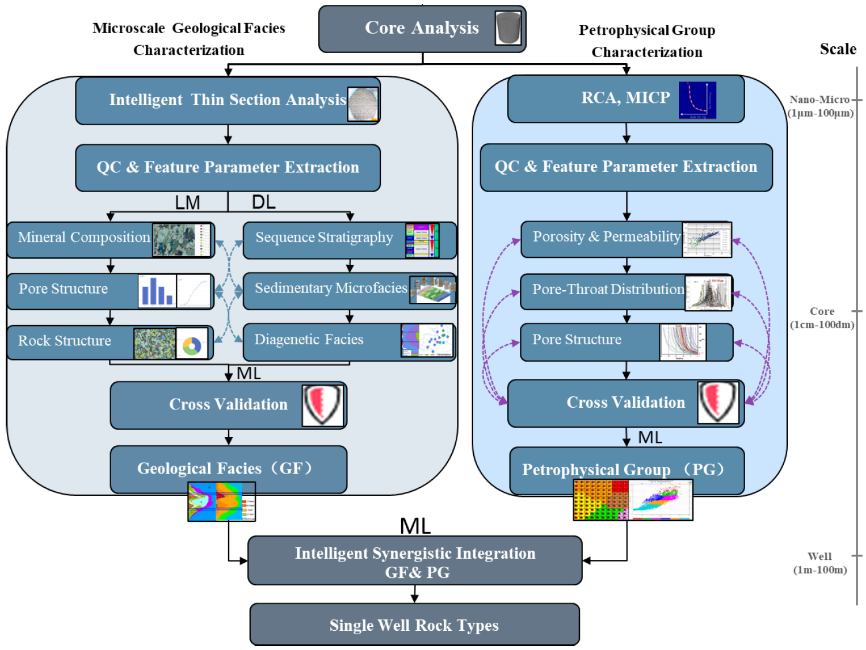

Figure 13—Integrated multi-scale characterization workflow bridging microscopic minerals to macroscopic reservoir properties.

Microscopic to Mesoscopic Integration: Mineral composition heterogeneity (quartz: 35-45%, feldspar: 20-30%) directly controls pore structure development through differential dissolution-precipitation processes. Grain contact relationship variations (point contacts: 15-30%) establish local permeability characteristics, while diagenetic modification patterns create distinct microenvironmental conditions controlling fluid flow pathways.

Mesoscopic to Macroscopic Integration: AI grain size curves successfully quantify millimeter-centimeter scale heterogeneity, providing high-resolution constraints for digital core modeling. Pore structure characteristics (connectivity: 0.6-0.8) determine flow unit boundaries, cementation patterns control reservoir compartmentalization, and heterogeneity patterns influence sweep efficiency during field development operations.

Future Cross-Scale Integration through Machine Learning: The proposed integration framework enables the use of machine learning algorithms to establish quantitative relationships between:

- Microscopic mineral assemblages and macroscopic porosity-permeability trends
- Grain-scale textural parameters and reservoir-scale flow characteristics
- Diagenetic intensity indices and production performance indicators

This multi-scale characterization approach has the potential to provide quantitative frameworks for:

- Reservoir quality prediction in unsampled intervals
- Optimal well placement and completion design
- Enhanced oil recovery strategy optimization
- Geological model parameterization and uncertainty quantification

The methodology demonstrates significant advancement in marine sandstone reservoir characterization, bridging the scale gap between detailed microscopic observations and practical reservoir development applications, ultimately enabling more efficient and sustainable hydrocarbon recovery from complex marine depositional systems.

Conclusions and Future Prospects

This study presents an innovative AI-driven workflow for intelligent thin section analysis utilizing the CoreSAM deep learning model, achieving significant advances in quantitative multi-scale heterogeneity characterization of marine sandstone reservoirs. Through systematic analysis of four representative wells with distinct sedimentary sequences, the technology accurately identifies complex marine bedding structures and quantitatively characterizes spatial variability in key reservoir parameters. The intelligent system demonstrates superior analytical capabilities by detecting previously unrecognized millimeter-scale heterogeneities within apparently uniform sedimentary structures, providing unprecedented resolution in reservoir characterization. Cross-directional anisotropy analysis reveals systematic differences between vertical and horizontal sample orientations, with vertical sections showing enhanced porosity but reduced connectivity compared to horizontal sections. The AI-enhanced quantification of compaction indices (0.1-1.4) and areal porosity distributions (1-15%) establishes preliminary frameworks for correlating depositional architecture with diagenetic modification patterns. The comprehensive "sample submission-intelligent identification-report generation" pipeline demonstrates exceptional promise for future development of integrated machine learning algorithms and intelligent geological modeling systems.

The authors acknowledge the support of PetroChina Research Institute of Petroleum Exploration and Development for providing research facilities and marine sandstone core samples. Special recognition goes to the Artificial Intelligence Research Center for technical support and algorithm development.

Nomenclature

AI = Artificial Intelligence
CNN = Convolutional Neural Network
CV = Computer Vision
DL = Deep Learning
HTC = Hybrid Task Cascade
R-CNN = Region-based Convolutional Neural Network
SAM = Segment Anything Model
SEM = Scanning Electron Microscopy
SVM = Support Vector Machine

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